

3. Let the ten's and unit's digits be x and y respectively, where $x > y$.
I. gives, $x - y = 5$... (i)
II. gives, $x + y = 7$... (ii)
 Solving (i) and (ii), we get : $x = 6$ and $y = 1$. So, the required number is 61.
 Thus, both I and II together give the answer.
 $\therefore$ The correct answer is (e).
4. Let the required number be x .
I. $\frac{4}{5}x - \frac{3}{4}x = 2.5 \Leftrightarrow \frac{x}{20} = 2.5 \Leftrightarrow x = 50$.
 So, $\frac{4}{5}x = \left(\frac{4}{5} \times 50\right) = 40$.
II. $x + \frac{x}{2} = 75 \Leftrightarrow \frac{3x}{2} = 75 \Leftrightarrow x = \left(\frac{75 \times 2}{3}\right) = 50$.
 So, $\frac{4}{5}x = \left(\frac{4}{5} \times 50\right) = 40$.
 Thus, either I or II alone gives the answer.
 $\therefore$ The correct answer is (c).
5. **II.** Let the greater number be x . Then, smaller number = $(30 - x)$.
I. $x - (30 - x) = \frac{x}{3} \Leftrightarrow 2x - 30 = \frac{x}{3} \Leftrightarrow 6x - 90 = x$
 $\Leftrightarrow 5x = 90 \Leftrightarrow x = 18$.
 So, smaller number = $30 - 18 = 12$.
 Thus, both I and II together give the answer.
 $\therefore$ The correct answer is (e).
6. Let the ten's and unit's digits be x and y respectively.
I. $x + y = 6$... (i)
II. $x = 2y$... (ii)
 Solving (i) and (ii), we get : $x = 4$, $y = 2$. So, the required number is 42.
 Thus, both I and II together give the answer.
 $\therefore$ The correct answer is (e).
7. **I.** Let the ten's and unit's digits of the number be a and b respectively.
 Then, $a + b = 8$.
II. Let the required number be x .
 Then, $\frac{x}{5} = \left(\frac{1}{2} \text{ of } 44\right) - 15 = 7 \Leftrightarrow x = 35$.
 So, the required number is 35.
 Thus, II alone gives the answer.
 $\therefore$ The correct answer is (b).
8. Let the required number be x .
I. $\frac{x}{2} - \frac{2x}{5} = 20 \Leftrightarrow \frac{5x - 4x}{10} = 20 \Leftrightarrow \frac{x}{10} = 20 \Leftrightarrow x = 200$.
II. $\frac{x}{4} = 25\% \text{ of } x \Leftrightarrow \frac{x}{4} = \frac{x}{4}$.
 Thus, I alone gives the answer.
 $\therefore$ The correct answer is (a).
9. **II.** Let the two numbers be $3x$ and $4x$.
I. $(3x)^2 = 9 \times 4x \Leftrightarrow 9x^2 = 36x \Leftrightarrow x = 4$.
 So, the numbers are 12 and 16.
 $\therefore$ Required difference = $16 - 12 = 4$.
 Thus, both I and II together give the answer.
 $\therefore$ The correct answer is (e).
10. Let the 8 consecutive real numbers be $x, (x + 1), (x + 2), (x + 3), (x + 4), (x + 5), (x + 6)$ and $(x + 7)$.
I. gives, $x(x + 1)(x + 2)(x + 3)(x + 4)(x + 5)(x + 6)(x + 7) = 34459425$.
 This equation cannot be solved for x .
II. gives, $x + (x + 1) + (x + 2) + (x + 3) + (x + 4) + (x + 5) + (x + 6) + (x + 7) = 84$
 $\Leftrightarrow 8x + 28 = 84 \Leftrightarrow 8x = 56 \Leftrightarrow x = 7$.
 $\therefore$ Third number = $x + 2 = 7 + 2 = 9$.
 Thus, II alone gives the answer.
 $\therefore$ The correct answer is (b).
11. Let the two numbers be x and y .
I. gives, $x + y = 2(x - y) \Leftrightarrow x = 3y$
 $\Leftrightarrow \frac{x}{y} = \frac{3}{1} \Leftrightarrow x : y = 3 : 1$.
 Thus, I only gives the answer.
II. does not give the answer.
 $\therefore$ Correct answer is (a).
12. The only two-digit number which is a multiple of 51, is 51. So, I alone gives the answer while II alone does not.
 $\therefore$ The correct answer is (a).
13. Let the ten's digit be x and the unit's digit be y .
 Then, number = $10x + y$.
I. $x + y = 13 \Leftrightarrow y = (13 - x)$.
 So, number = $10x + (13 - x)$.
II. $[10x + (13 - x)] - [10(13 - x) + x] = 45$
 $\Leftrightarrow (9x + 13) - (130 - 9x) = 45$
 $\Leftrightarrow 18x = 162 \Leftrightarrow x = 9$.
 $y = 13 - x = 13 - 9 = 4$.
 So, required number = 94.
 Thus, both I and II together give the answer.
 $\therefore$ The correct answer is (e).
14. Let the ten's and unit's digits be x and y respectively. Then,
I. $x + y = 10$ and $\frac{x}{y} = \frac{1}{4}$.
II. $xy = 16$ and $\frac{x}{y} = \frac{4}{1}$.
 $\therefore$ II gives, $x^2 = 64 \Leftrightarrow x = 8$.
 So, $4y = 8 \Leftrightarrow y = 2$.
 Thus, II alone gives the answer while I alone does not.
 $\therefore$ The correct answer is (b).
15. Let the ten's digit be x and the unit's digit be y .
II. $y > x$.
I. $xy = 72$ and $y - x = 1$ or $y = x + 1$
 $\Leftrightarrow x(x + 1) = 72 \Leftrightarrow x^2 + x - 72 = 0$
 $\Leftrightarrow x^2 + 9x - 8x - 72 = 0$
 $\Leftrightarrow x(x + 9) - 8(x + 9) = 0$
 $\Leftrightarrow (x + 9)(x - 8) = 0 \Leftrightarrow x = 8$.
 So, $y = x + 1 = 9$
 $\therefore$ Required number = 89.
 Thus, both I and II together give the answer.
 $\therefore$ The correct answer is (e).
16. Let the ten's and unit's digit be x and y respectively.
I. $x + y = 7$.